Research Statement

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My research goal is to transform complex genomic data into meaningful insights that clinicians and biologists can use in research and patient care. I build computational methods that discover cancer biomarkers and support treatment decisions. I pursue this goal by combining deep-learning models, statistical genomics, and explainable artificial intelligence (AI). Each method is designed to address a specific clinical question. The research methods I develop and apply can be decomposed into the following types.

1. Development of **Deep Learning** pipelines to address the high dimensionality and sparsity of heterogeneous cancer data, including gene expression, somatic variant, whole-genome sequencing, and whole-exome sequencing data.
2. Application of **Statistical** methodologies to identify genetic variant-level biomarkers associated with aggressive prostate cancer.
3. Use of a **Similarity-matching algorithm** to translate discovered biomarkers into drug candidates through drug repurposing.
4. Use of **Explainable AI** approaches so that model-derived biomarkers and predictions are interpretable to biologists and clinicians.
5. Development of a **Generative AI** pipeline that combines variant-aware literature retrieval, ClinVar annotations, and large language model based summarization to explain the characteristics of variants.

Research Projects

- **Multi-cancer drug repurposing framework:** To identify common drug candidates for breast and prostate cancer, I developed a drug repurposing framework using a similarity-matching algorithm [1]. The framework takes patient gene expression and drug response data as input and produces an output score for each drug based on its relevance to the affected genes. Since cancer is heterogeneous, we used mutation data (VCF) to create more homogeneous patient groups. The results revealed that analyzing biomarker-defined patient subgroups produced more relevant drug candidates than analyzing all patients together and identified Mitoxantrone and Genistein as promising drugs. Using normalized discounted cumulative gain (NDCG), I validated the rankings.
- **A statistical approach to identify variants unique to aggressive prostate cancer:** Identifying variants associated with aggressive prostate cancer can help explain why some tumors progress more rapidly. I developed a hypothesis-testing framework to compare variants between aggressive and indolent prostate cancer patients [2]. The analysis identified four significant variants, including a previously unreported variant affecting the TRPM7 gene. Moreover, I performed pathway analysis which showed that these variants influence cancer-related pathways, including PI3K–Akt signaling. In an independent dataset, I found that the proposed variants achieved 97% classification accuracy, compared with 65% for randomly selected variants.
- **Overcoming data scarcity in aggressive prostate cancer biomarker discovery:** The limited number of aggressive prostate cancer cases in genomic datasets makes reliable variant-level biomarker discovery difficult. To address this limitation, I developed CLOVER, a cross-cancer deep-learning framework that integrates somatic variant data from five biologically related cancers [3]. CLOVER combines an autoencoder that compresses the high-dimensional variant data with a supervised multiclass classifier that jointly predicts cancer type and disease aggressiveness. The model achieved 87% balanced accuracy and substantially outperformed a model trained only on prostate cancer data. Using SHAP-based explainability and pathway analysis, I also identified and biologically validated prostate-specific and cross-cancer biomarkers.

- **Addressing the sparsity problem in somatic mutation data analysis:** Somatic mutation data are highly sparse because each patient carries mutations at only a small fraction of all possible genomic positions, making it difficult for conventional models to learn reliable patterns. To address this problem, I developed a graph-based framework that represents patients through similarity relationships and combines a variational graph autoencoder with contrastive learning. The model learns informative patient representations despite sparse inputs. Furthermore, I used GNN explainer that identifies the variants driving each prediction. This research paper is currently under review in the "IEEE Transactions on Computational Biology and Bioinformatics" journal.
- **Network-based biomarker and pathway identification for viral and chronic disease:** In earlier work, I used network-based bioinformatics to identify shared molecular mechanisms across risk factor diseases of SARS-CoV-2 virus. Across studies, I analyzed public gene expression data to identify common genes associated with COVID-19 comorbidities. I constructed protein–protein interaction and gene-regulatory networks and applied network and pathway analyses to identify important genes and biological processes. Together, this work demonstrated how computational analysis of public biological data can reveal shared disease mechanisms, potential biomarkers, and candidate treatments [4, 5].

Ongoing Projects and Future Research Directions

Across my Ph.D. research, I have worked at the intersection of deep learning, statistical genomics, and explainable AI applied to oncology. That experience has prepared me to pursue the following research directions.

- **Generative AI assistant for mutation interpretation:** Clinical information about genetic variants is widely scattered across public databases and scientific literature. As a result, researchers without a biological background face challenges in interpreting variants and conducting computational biology research. To address this problem, I am developing a generative-AI assistant that retrieves variant-specific literature and ClinVar annotations, and uses a large language-model to summarize the characteristics of variants. This bridges automated modeling and the narrative reasoning clinicians rely on.
- **Multi-task learning system for single-cell biomarker discovery for Prostate cancer:** Bulk RNA sequencing data generally hide important differences among individual tumor cells, which can be solved using single-cell RNA sequencing data. After reviewing existing single-cell prostate cancer studies, I found that most current pipelines analyze cell types, disease states, regulatory networks, and pathway activity separately. To address this gap, I will develop a graph-based multi-task representation learning system that integrates single-cell RNA-sequencing data with gene regulatory network information. The model will learn shared cell representations and use task-specific components for cell-type classification, cancer-state prediction, cell-state transition analysis, and pathway-activity scoring. The Department of Defense (DoD) Prostate Cancer Research Program (PCRP) is a target funding program for this research.
- **Multimodal foundation model for cancer-variant interpretation:** Existing biological foundation models generally analyze DNA, RNA, or protein sequences separately, even though the functional effect of a variant may involve all three modalities. I plan to develop a multimodal foundation model that connects pretrained DNA, RNA, and protein encoders through cross-attention layers. The model will generate a unified representation of each variant and use task-specific prediction heads to estimate its effect on gene expression, protein function, and cancer aggressiveness. The NIH/NCI Informatics Technology for Cancer Research (ITCR) is a target funding program for this research.

- **Predicting treatment-induced transformation in prostate cancer:** Some prostate tumors escape hormone therapy by changing into more treatment-resistant cell states, but the drivers of this transformation remain unclear. I will develop a dynamic systems model using longitudinal single-cell, spatial, and treatment-response data to identify the interactions that trigger this change. The model will predict biomarkers and drug combinations that may prevent treatment resistance. The NIH/NCI Cancer Systems Biology Consortium (CSBC) is a target funding program for this research.
- **Privacy-preserving biomarker discovery across hospitals:** Rare aggressive cancers are difficult to study because genomic datasets are divided across hospitals and cannot always be shared. I will develop a federated learning framework that learns from multiple institutions without transferring patient-level data. The model will address differences among hospitals and identify reliable biomarkers across diverse populations. The NSF Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science (SCH) is a target funding program for this research.

References

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